



Qualification Pack

Machine Operator & Programmer CNC Milling

QP Code: CPC/CAP/Q0801

Version: 1.0

NSQF Level: 4

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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CPC/CAP/Q0801: Machine Operator & Programmer CNC Milling

Brief Job Description

Machine Operator & Programmer CNC Milling Qualification is designed to familiarize individuals with the CNC milling process and provide opportunities to work in the tooling industry

Personal Attributes

Technical aptitude, Safety-conscious, Reliability and punctuality, Organizational skills, Problem-solving ability, Teamwork and communication, Quality-focused mindset, Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CPC/CAP/N0801: Perform machining operations on metal or plastic material using Conventional Milling machine](#)
2. [CPC/CAP/N0803: Programming of Computer Numerically Controlled \(CNC\) Milling Machines](#)
3. [CPC/CAP/N0802: Perform machining operations on metal or plastic workpieces using Computer Numerically Controlled Milling machines](#)
4. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
5. [CPC/CAP/N0405: To Effective working with others](#)
6. [CPC/CAP/N0404: Computer and data entry in MS Office/Office Open Source software](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
Sub-Sector	Petrochemicals
Occupation	CNC Milling
Country	India
NSQF Level	4

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Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.5001
Minimum Educational Qualification & Experience	12th Class (or Equivalent) OR 10th Class (or Equivalent) with 2 Years of experience NTC/NAC OR 11th Class (or Equivalent) with 1.5 years of experience relevant experience OR 10th Class (or Equivalent) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-04-CP-04127-2025-V2-CIPET
NQR Version	2.0

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CPC/CAP/N0801: Perform machining operations on metal or plastic material using Conventional Milling machine

Description

This unit covers performing milling operations such as Plain Milling, Face Milling, Side Milling, Angular Milling, Profile Milling, End Milling etc

Scope

The scope covers the following :

- This unit covers the following:
- Understanding the working principle & construction of milling machine
- Working safely
- Carrying out operations on conventional milling machine
- Measuring and checking the workpiece as per specification

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** understand and comply with safety, environmental & other relevant regulations and guidelines
- PC2.** wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing Milling operations
- PC3.** ensure work area is clean and safe
- PC4.** ensure that machine safety guards are in place and are in correctly working condition
- PC5.** ensure that all tools, equipments are in a safe and usable conditions

Prerequisite for operating Conventional milling machine

To be competent, the user/individual on the job must be able to:

- PC6.** ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card
- PC7.** read & understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances
- PC8.** check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size
- PC9.** plan to perform the machining operations and the sequence of operations as per required job specifications on milling machine
- PC10.** obtain all the appropriate tools and measuring instruments/ gauges required for the job
- PC11.** check the milling machine for its functioning and ensure that it is ready for operation
- PC12.** prepare the milling machine for the operations by mounting and setting the required work holding devices, tool holding devices and cutting tools

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PC13. clarify any doubt, if any and see necessary instruction /training on the operation of the machine whenever required

PC14. hold the work piece securely and correctly, without distortion

PC15. adjust the machine settings as per job requirement to maintain desired accuracy

Performing Operations on conventional Milling machine

To be competent, the user/individual on the job must be able to:

PC16. adjust and set the speed and feed of the milling machine to achieve the job specifications

PC17. operate the machine tool controls safely and correctly, in line with operational procedures both in manual and power modes

PC18. stop the milling machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after and emergency

PC19. should be able to use the milling machine accessories and attachments such as Universal Milling attachment, Slotting attachment, Circular milling attachment , Dividing head etc.

PC20. • perform various milling operations using different tools to produce components with various features.
• Tool: H.S.S tools, brazed carbide tip tools, interchangeable carbide insert tools, end mill cutter, side milling cutter, T slot cutter, drills, reamers, boring tool etc.
• Component Feature: facing, slotting, pocket milling, circular pocket, drilling, reaming, boring, counter boring, profile milling keyways, grooves, slots, gear cutting, helical milling or special profiles

PC21. produce components as per required quality standards and free from burrs & sharp edges

PC22. shall achieve given production targets

PC23. shall be able to apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy

PC24. shall be able to use coolants/ cutting fluids for different combinations of work piece and tool as per different locations

PC25. shall be able to observe and report any difficulties/ discrepancies that may arise during the machine operation and carry out the corrective actions as per instructions

PC26. correctly shutting down the machine on completion of the machining operations, removing and disposing of the chips/ waste and critical parameters different locations

PC27. use of measuring instruments/ gauges to check the critical parameters

PC28. shall be able to carry out the corrective action, in the case of deviation from the required specifications

PC29. report the problem to the supervisor, if it cannot be resolved.

PC30. seek guidance from the supervisor/ specialist if the problem is outside his/her area of competence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. general policies, procedures rules & regulations followed in company

KU2. employment terms & conditions, entitlements, job role and responsibilities

KU3. Do and Do not to be followed in the company

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- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** relevant people and their responsibilities within the work area
- KU6.** work related procedures and documentation and their importance
- KU7.** quality management related documentation, if any and their importance
- KU8.** working principle & construction of conventional milling machine
- KU9.** start, stop, emergency buttons and machine controls
- KU10.** safety mechanisms on the machine, safety guards and procedure to check their functionality
- KU11.** hazards associated with the milling operations and safety to be observed
- KU12.** meaning and purpose of machining operations on milling machine
- KU13.** understanding and use of the metric system of measurements
- KU14.** concept of engineering drawing, isometric and orthographic projection, sectional views, auxiliary views, dimensioning
- KU15.** understanding of the geometric tolerances, Hole and shaft basis of ISO tolerance, straightness, flatness, circularity, ovality, surface finish and their symbols
- KU16.** type of tools for various milling machine operations Plain Milling, Face Milling, Side Milling, Angular Milling, Profile Milling, End Milling etc
- KU17.** type of tool materials- classification, properties and their application
- KU18.** understanding of cutting parameters and their selection i.e. cutting speed, feed, depth of cut and their effect on tool life and surface finish
- KU19.** type of coolants/ cutting fluids, classification, application and effect on tool life and surface finish
- KU20.** understanding of the milling machine accessories, attachments and their uses
- KU21.** types of various work holding & tool holding devices and their application
- KU22.** understanding error messages on machine and taking appropriate corrective action
- KU23.** importance of securing the work-piece/raw material correctly using appropriate devices and mechanisms
- KU24.** how to check the quality of machined components against the specified quality standards

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & interpret correctly the job specifications from drawing/ job card, manuals, safety instructions etc. in English and/or local language
- GS2.** able to fill up the required formats/ documents in English and / or local language
- GS3.** interact and communicate with supervisor or other company personnel as per requirement
- GS4.** perform simple numerical computation such as addition, subtraction, multiplication, division, fractions and decimal, percentages and proportions, simple ratios and average
- GS5.** check and clarify task-related information
- GS6.** plan, prioritize and sequence work operations as per job requirements
- GS7.** shall be able to detect out of tolerance limit of component or any malfunctioning of the machine and take corrective action



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- GS8.** decide when to contact supervisor in case of any unresolved problems
- GS9.** analyse and interpret geometric dimensions and tolerances and apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5	10	-	-
PC1. understand and comply with safety, environmental & other relevant regulations and guidelines	1	2	-	-
PC2. wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing Milling operations	1	2	-	-
PC3. ensure work area is clean and safe	1	2	-	-
PC4. ensure that machine safety guards are in place and are in correctly working condition	1	2	-	-
PC5. ensure that all tools, equipments are in a safe and usable conditions	1	2	-	-
<i>Prerequisite for operating Conventional milling machine</i>	15	25	-	-
PC6. ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card	1	2	-	-
PC7. read & understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances	1	2	-	-
PC8. check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size	1	2	-	-
PC9. plan to perform the machining operations and the sequence of operations as per required job specifications on milling machine	1	2	-	-
PC10. obtain all the appropriate tools and measuring instruments/ gauges required for the job	1	2	-	-
PC11. check the milling machine for its functioning and ensure that it is ready for operation	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. prepare the milling machine for the operations by mounting and setting the required work holding devices, tool holding devices and cutting tools	2	3	-	-
PC13. clarify any doubt, if any and see necessary instruction /training on the operation of the machine whenever required	2	3	-	-
PC14. hold the work piece securely and correctly, without distortion	2	3	-	-
PC15. adjust the machine settings as per job requirement to maintain desired accuracy	2	3	-	-
<i>Performing Operations on conventional Milling machine</i>	30	45	-	-
PC16. adjust and set the speed and feed of the milling machine to achieve the job specifications	2	3	-	-
PC17. operate the machine tool controls safely and correctly, in line with operational procedures both in manual and power modes	2	3	-	-
PC18. stop the milling machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after and emergency	2	3	-	-
PC19. should be able to use the milling machine accessories and attachments such as Universal Milling attachment, Slotting attachment, Circular milling attachment , Dividing head etc.	2	3	-	-
PC20. - perform various milling operations using different tools to produce components with various features. - Tool: H.S.S tools, brazed carbide tip tools, interchangeable carbide insert tools, end mill cutter, side milling cutter, T slot cutter, drills, reamers, boring tool etc. - Component Feature: facing, slotting, pocket milling, circular pocket, drilling, reaming, boring, counter boring, profile milling keyways, grooves, slots, gear cutting, helical milling or special profiles	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. produce components as per required quality standards and free from burrs & sharp edges	2	3	-	-
PC22. shall achieve given production targets	2	3	-	-
PC23. shall be able to apply roughing and finishing cuts, considering the effect on tool life, surface finish and dimensional accuracy	2	3	-	-
PC24. shall be able to use coolants/ cutting fluids for different combinations of work piece and tool as per different locations	2	3	-	-
PC25. shall be able to observe and report any difficulties/ discrepancies that may arise during the machine operation and carry out the corrective actions as per instructions	2	3	-	-
PC26. correctly shutting down the machine on completion of the machining operations, removing and disposing of the chips/ waste and critical parameters different locations	2	3	-	-
PC27. use of measuring instruments/ gauges to check the critical parameters	2	3	-	-
PC28. shall be able to carry out the corrective action, in the case of deviation from the required specifications	2	3	-	-
PC29. report the problem to the supervisor, if it cannot be resolved.	2	3	-	-
PC30. seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence	2	3	-	-
NOS Total	50	80	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0801
NOS Name	Perform machining operations on metal or plastic material using Conventional Milling machine
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Milling, CNC Milling
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0803: Programming of Computer Numerically Controlled (CNC) Milling Machines

Description

This NOS covers making programs for Computer Numerically Controlled (CNC) Milling machines/ machining centers. Programming can be done manually or using appropriate CAM software. The program is transferred to the machine's controller by entering it at the console, transmitting it through a wired link, or copying it through a data storage device. The candidate will be expected to perform safe operations with a minimum of supervision, taking personal responsibility for one own actions and for the quality and accuracy of the work produced

Scope

The scope covers the following :

- This unit/ task covers the following:
- Working safely
- Preparing for programming CNC milling machine for production
- Carrying out programming for CNC milling machine
- Test and prove the program on the CNC milling Machine

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.**
 - adhere to procedures and guidelines for personal protective equipment (PPE)
 - and other relevant safety regulations while programming CNC milling machines
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that machine guards are in place and are correctly adjusted
- PC5.** read and understand safety instructions, warning signs on the machine
- PC6.** ensure work area is clean and safe from hazards
- PC7.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition

Preparing for programming CNC Milling machine for production

To be competent, the user/individual on the job must be able to:

- PC8.** ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card
- PC9.** read and establish job requirements from the job specification document accurately
- PC10.** follow job instructions, assembly drawings and laid down procedures at all times
- PC11.** report and rectify incorrect and inconsistent information in job specification documents as per organization procedures

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- PC12.** use and extract information from reference charts, tables, graphs and standards
- PC13.** prepare the work area as per procedure or operational specification
- PC14.** conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly
- PC15.** determine what operational objectives and targets need to be achieved and how best the machine needs to be programmed to achieve this CNC programming operations: preparing, loading, storing in appropriate format, proving the part program, trial runs
- PC16.** extract and use information from engineering drawings and related specifications in relation to work undertaken
- PC17.** identify tool requirements from tooling layout and assess their suitability
- PC18.** identify suitable workholding or fixturing device as per the job requirement
- PC19.** ensure the correct and latest part-program is uploaded onto the CNC system

Carrying out programming for CNC Milling machine

To be competent, the user/individual on the job must be able to:

- PC20.** make the CNC program with commands for tool motions, spindle motions, miscellaneous functions and tool change, in syntax corresponding to the machine and control system on which the component will be machined
- PC21.** various ways to make CNC program are by writing it on paper or in a computer's text editor, or using CAM software or controllers on machine Ways: written, directly entered into the machine controller, using computer software- CAM software
- PC22.**
 - ensure that the part program is efficient and results in minimal cycle time, with
 - optimal cutting parameters and no unnecessary tool motions
- PC23.** use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine
- PC24.** transfer the program to the machine by entering it at the console or transmitting it through a wired link or through a data transfer device
- PC25.** follow the correct procedures for calling up the program and dealing with any error messages or faults
- PC26.** handle the typical problems that can occur with the programming, loading and editing activities effectively using approved procedures
- PC27.** save the proven program in the appropriate storage medium – paper, computer hard disk, etc. - and location
- PC28.** complete relevant documentation as per organizational procedure
- PC29.** leave the work area in a safe and tidy condition on completion of the activities

Test and prove the program on the CNC Milling Machine

To be competent, the user/individual on the job must be able to:

- PC30.** obtain appropriate equipment or tools needed as per job requirements
- PC31.** ensure that all measuring equipment is calibrated and approved for usage
- PC32.** ensure that the tools and fixtures are in usable condition(eg. free from breakage, damage, calibration, etc.)
- PC33.** pre-set the tooling appropriately using setting jigs/fixtures
- PC34.** seek any necessary instruction/training on the operation of the machine where required
- PC35.** mount tools in the correct positions in the tool turret or magazine

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- PC36.** check that the tools have been mounted in positions corresponding to tool numbers in the part program
- PC37.** mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces.
- PC38.** enter work offset and tool data on the machine – X and Z offsets, tool orientation and LOe radius for lathes; length offsets and tool radius for machining centers
- PC39.** ensure that tool data has been entered in offset number corresponding to the tool offset numbers in the part program
- PC40.** deal with error messages and faults on the program or equipment
- PC41.** cut a trial part using single block run, dry run and feed and speed override controls
- PC42.** edit the program and adjust tool and wear offsets to correct any dimensional errors on the part
- PC43.** ensure that the trial part conforms to drawing specifications in terms of dimensions, surface finishes and geometrical parameters like concentricity, parallelism, runout, etc.
- PC44.** correct the tool wear offsets whenever required, based on the results of the period inspection
- PC45.** change worn out tools and indexable inserts whenever required
- PC46.** after every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets
- PC47.** return worn out cutting tools, workholding device / fixtures / instruments / drawings to store
- PC48.** ensure that there is no damage to the tool/fixture while doing the prove-out
- PC49.** shut down the equipment to a safe condition on conclusion of the activities
- PC50.** deal promptly and effectively with problems within span of responsibility and control and report those that cannot be solved

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** deal promptly and effectively with problems within span of responsibility and control and report those that cannot be solved
- KU2.** employment terms & conditions, entitlements, job role and responsibilities
- KU3.** Do and Do not to be followed in the company
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** relevant people and their responsibilities within the work area
- KU6.** work related procedures and documentation & their importance
- KU7.** quality management related documentation, if any & their importance
- KU8.** specific safe working practices, CNC programming procedures and environmental regulations that must be observed
- KU9.** hazards associated with carrying out the machining operations on a CNC machine and how can they be minimized
- KU10.** personal protective equipment to be used during the machining activities on a CNC machine and where can it be obtained
- KU11.** safety mechanism on the machine and how to check if they are functioning properly

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- KU12.** types and sources of appropriate job specifications
- KU13.** common terminology used in CNC programming features of produced CNC program
- KU14.** selection of strategies based on material and fixturing, holding and clamping force
- KU15.** the factors which will determine selection and use of tungsten carbide and tips Factors hardness of the component material machinability characteristics of the material tolerances to be achieved surface finish to be achieved geometrical accuracies like ovality straightness and flatness to be achieved rigidity of work holding
- KU16.** importance of tool selection based on material, finish required and tolerances achieved
- KU17.** importance of cutter engagement and exit
- KU18.** factors affecting tool life and importance and effect of the depth of cut, RPM and feed
- KU19.** how to read and interpret first and third angle component drawings
- KU20.** how to extract information from engineering drawings or data and related specifications
- KU21.** how to use the function keys and user interface of the machine control system
- KU22.** determination and entry of work and tool offsets, tool wear data
- KU23.** main features and working parts of the CNC machine, and the accessories that can be used
- KU24.** importance of following specified machining sequences and procedures
- KU25.** importance and procedures to ensure that tools and equipment are in a safe and usable condition
- KU26.** various CNC operations that can be performed, and the methods and equipment used
- KU27.** methods of setting the work-holding devices, and the tools and equipment that can be used
- KU28.** various tool holding devices that are used, and the methods of correctly mounting and securing the cutting tools to the tool holders
- KU29.** how to set the machine controller in the program and editing mode, and enter or download the prepared program
- KU30.** function of error messages, and appropriate subsequent action
- KU31.** importance of proving the program, how to do it and selecting the correct proving tools
- KU32.** need for storing program tapes and disks safely and correctly, away from contaminants and electromagnetic sources
- KU33.** quality control procedures that are used, inspection checks to be carried out, and the equipment that will need to be used
- KU34.** importance to report problems in a timely manner
- KU35.** importance of writing programs that are easily editable or correctable by the next person
- KU36.** methods of checking quality of the shaped components against the required quality standards

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret correctly the job specifications from growing/ job card, manuals, safety instructions etc. in English and/ or local language
- GS2.** able to fill up the required formats/ documents in English and / or local language



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- GS3.** interact and communicate with supervisor or other company personnel as per requirement
- GS4.** shall be able to use simple numerical computation such as addition, subtraction, multiplication, division, fractions and decimal, percentages and proportions, simple ratios and average
- GS5.** check and clarify task-related information
- GS6.** plan, prioritize and sequence work operations as per job requirements
- GS7.** shall be able to detect out of tolerance limit of component or any malfunctioning of the machine and take corrective action
- GS8.** shall be able to decide when to contact supervisor in case of any unresolved problems
- GS9.** shall be able to analyse and interpret geometric dimensions & tolerance & apply balanced judgments to different instructions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	7	21	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	3	-	-
PC2. • adhere to procedures and guidelines for personal protective equipment (PPE) • and other relevant safety regulations while programming CNC milling machines	1	3	-	-
PC3. work following laid down procedures and instructions	1	3	-	-
PC4. ensure that machine guards are in place and are correctly adjusted	1	3	-	-
PC5. read and understand safety instructions, warning signs on the machine	1	3	-	-
PC6. ensure work area is clean and safe from hazards	1	3	-	-
PC7. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	1	3	-	-
<i>Preparing for programming CNC Milling machine for production</i>	12	36	-	-
PC8. ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card	1	3	-	-
PC9. read and establish job requirements from the job specification document accurately	1	3	-	-
PC10. follow job instructions, assembly drawings and laid down procedures at all times	1	3	-	-
PC11. report and rectify incorrect and inconsistent information in job specification documents as per organization procedures	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use and extract information from reference charts, tables, graphs and standards	1	3	-	-
PC13. prepare the work area as per procedure or operational specification	1	3	-	-
PC14. conduct a preliminary check of the readiness of the program so that the CNC machine operates correctly	1	3	-	-
PC15. determine what operational objectives and targets need to be achieved and how best the machine needs to be programmed to achieve this CNC programming operations: preparing, loading, storing in appropriate format, proving the part program, trial runs	1	3	-	-
PC16. extract and use information from engineering drawings and related specifications in relation to work undertaken	1	3	-	-
PC17. identify tool requirements from tooling layout and assess their suitability	1	3	-	-
PC18. identify suitable workholding or fixturing device as per the job requirement	1	3	-	-
PC19. ensure the correct and latest part-program is uploaded onto the CNC system	1	3	-	-
<i>Carrying out programming for CNC Milling machine</i>	10	30	-	-
PC20. make the CNC program with commands for tool motions, spindle motions, miscellaneous functions and tool change, in syntax corresponding to the machine and control system on which the component will be machined	1	3	-	-
PC21. various ways to make CNC program are by writing it on paper or in a computer's text editor, or using CAM software or controllers on machine Ways: written, directly entered into the machine controller, using computer software- CAM software	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. • ensure that the part program is efficient and results in minimal cycle time, with • optimal cutting parameters and no unnecessary tool motions	1	3	-	-
PC23. use subprograms and canned cycles, to reduce program size and input time and avoid memory overflow on the machine	1	3	-	-
PC24. transfer the program to the machine by entering it at the console or transmitting it through a wired link or through a data transfer device	1	3	-	-
PC25. follow the correct procedures for calling up the program and dealing with any error messages or faults	1	3	-	-
PC26. handle the typical problems that can occur with the programming, loading and editing activities effectively using approved procedures	1	3	-	-
PC27. save the proven program in the appropriate storage medium – paper, computer hard disk, etc. - and location	1	3	-	-
PC28. complete relevant documentation as per organizational procedure	1	3	-	-
PC29. leave the work area in a safe and tidy condition on completion of the activities	1	3	-	-
<i>Test and prove the program on the CNC Milling Machine</i>	21	63	-	-
PC30. obtain appropriate equipment or tools needed as per job requirements	1	3	-	-
PC31. ensure that all measuring equipment is calibrated and approved for usage	1	3	-	-
PC32. ensure that the tools and fixtures are in usable condition(eg. free from breakage, damage, calibration, etc.)	1	3	-	-
PC33. pre-set the tooling appropriately using setting jigs/fixtures	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. seek any necessary instruction/training on the operation of the machine where required	1	3	-	-
PC35. mount tools in the correct positions in the tool turret or magazine	1	3	-	-
PC36. check that the tools have been mounted in positions corresponding to tool numbers in the part program	1	3	-	-
PC37. mount the part on machine firmly in the specified work holding devices, with the appropriate clamping forces.	1	3	-	-
PC38. enter work offset and tool data on the machine – X and Z offsets, tool orientation and LOe radius for lathes; length offsets and tool radius for machining centers	1	3	-	-
PC39. ensure that tool data has been entered in offset number corresponding to the tool offset numbers in the part program	1	3	-	-
PC40. deal with error messages and faults on the program or equipment	1	3	-	-
PC41. cut a trial part using single block run, dry run and feed and speed override controls	1	3	-	-
PC42. edit the program and adjust tool and wear offsets to correct any dimensional errors on the part	1	3	-	-
PC43. ensure that the trial part conforms to drawing specifications in terms of dimensions, surface finishes and geometrical parameters like concentricity, parallelism, runout, etc.	1	3	-	-
PC44. correct the tool wear offsets whenever required, based on the results of the period inspection	1	3	-	-
PC45. change worn out tools and indexable inserts whenever required	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC46. after every change of a worn out tool or insert, cut a trial part and correct any dimensional inaccuracies by adjusting the tool offsets or wear offsets	1	3	-	-
PC47. return worn out cutting tools, workholding device / fixtures / instruments / drawings to store	1	3	-	-
PC48. ensure that there is no damage to the tool/fixture while doing the prove-out	1	3	-	-
PC49. shut down the equipment to a safe condition on conclusion of the activities	1	3	-	-
PC50. deal promptly and effectively with problems within span of responsibility and control and report those that cannot be solved	1	3	-	-
NOS Total	50	150	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0803
NOS Name	Programming of Computer Numerically Controlled (CNC) Milling Machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Milling, CNC Milling
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0802: Perform machining operations on metal or plastic workpieces using Computer Numerically Controlled Milling machines

Description

This unit covers the operation of Computer Numerically Controlled (CNC) milling machine in order to perform machining operations on metal and plastic components, as per specifications provided. It includes machine setting and basic programming. This involves removal of material by machining from the work-piece

Scope

The scope covers the following :

- This unit/task covers the following:
- Understanding the working principle of CNC milling machine
- Working safely
- Carry out machining operations using CNC milling machine
- Measuring and checking the workpiece as per specifications

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** understand and comply with safety, environmental & other relevant regulations and guidelines
- PC2.** wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing milling operations
- PC3.** ensure work area is clean and safe
- PC4.** ensure that machine safety guards are in place and are in correctly working condition
- PC5.** ensure that all tools, equipments are in a safe and usable conditions

Prerequisite for operating CNC milling machine

To be competent, the user/individual on the job must be able to:

- PC6.** ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card
- PC7.** read and understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances
- PC8.** check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size
- PC9.** plan to perform the machining operations and the sequence of operations as per required job specifications on CNC milling machine
- PC10.** obtain all the appropriate tools and measuring instruments/ gauges required for the job
- PC11.** check the CNC milling machine for its functioning and ensure that it is ready for operation

Qualification Pack

- PC12.** prepare the CNC milling machine for the operations by mounting and setting the required workholding & tool holder devices and cutting tools
- PC13.** clarify any doubt, if any and see necessary instruction /training on the operation of the CNC milling machine whenever required
- PC14.** hold the work piece securely and correctly, without distortion
- PC15.** adjust the CNC milling machine settings as per job requirement to maintain desired accuracy
- PC16.** perform daily maintenance of machine according to defined checklist, at the beginning of day's shifts

Performing Operations on CNC milling machine

To be competent, the user/individual on the job must be able to:

- PC17.** use and extract information from engineering drawings, dimensioning and tolerances
- PC18.** use and extract information from reference charts, tables, graphs and Engineering standards
- PC19.** load and unload component(s) using pre-determined fixtures or work holding devices as per work instructions
- PC20.** make basic program and check correctness of program through dry run and single block check
- PC21.** adjust and set the speed and feed of the CNC milling machine to achieve the job specifications
- PC22.** operate the machine tool controls safely and correctly, in line with operational procedures
- PC23.** stop the CNC milling machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after the emergency
- PC24.** do first part cutting trial by setting tool offsets to get oversize part
- PC25.** measure the critical parameters of the machined component on machine (without removing from the machine), after the trial run
- PC26.** correct the offsets based on the measurements by accessing program edit facility in order to enter tooling data
- PC27.** measure the component after unloading to check for accuracy in the critical parameters as per job specifications
- PC28.** produce machined components that combine different machining operations and have a range of features
- PC29.** follow the specified machining sequence and procedure as per job specifications
- PC30.** interpret in-built machine alarms and respond to the same as per operating manual or specified instructions
- PC31.** observe for inconsistency in dimensions due to tool wear and correct the offsets accordingly
- PC32.** ensure that machine settings are adjusted as and when required, either by self or the setter, to maintain the required accuracy
- PC33.** identify when tools need replacement and replace worn tool with new tool
- PC34.** produce components as per required standards
- PC35.** report problems and seek appropriate assistance in a timely manner
- PC36.** complete documentation during and post operations as per organizational procedures and applicable quality management system

Qualification Pack

PC37. return the machine and all tools and equipment to the correct location on completion of activities

PC38. leave the work area in a safe and tidy condition on completion of job activities as per 5S practices

Seeking Guidance for unresolved Problems

To be competent, the user/individual on the job must be able to:

PC39. Report the problem to the supervisor, if it cannot be resolved

PC40. seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. general policies, procedures rules and regulations followed in the company

KU2. employment terms & conditions, entitlements, job role & responsibilities

KU3. Do and Do not to be followed in the company

KU4. reporting structure, inter-dependent functions, lines and procedures in the work area

KU5. relevant people and their responsibilities within the work area

KU6. work related procedures and documentation and their importance

KU7. quality management related documentation, if any and their importance

KU8. working principle and construction of the CNC milling machine

KU9. start, stop, emergency buttons and machine controls of CNC milling machine

KU10. safety mechanisms on the machine, safety guards and procedure to check their functionality

KU11. hazards associated with the CNC milling operations and safety to be observed

KU12. understanding and use of the metric system of measurements

KU13. absolute and incremental systems of tool positioning and offsetting

KU14. machine zero, work piece zero, work offsets, tool offsets

KU15. tool radius compensation- its necessity and effects of not using it

KU16. types and sources of appropriate job specifications

KU17. basic programming, canned cycles, G & M codes in CNC milling

KU18. tools and equipment used for machining operations on a CNC machines

KU19. various CNC milling operations that can be performed, and the methods and equipment used

KU20. correct techniques and procedures to carry out specific machining operations on a CNC milling

KU21. understanding error messages on machine and taking appropriate corrective action

KU22. importance of securing the work-piece/raw material correctly using appropriate devices and mechanisms and setting the work-holding device in relationship to the machine axis and reference points

KU23. common problems that can occur in CNC milling operations and their implications & correct procedures to address problems commonly encountered during CNC milling operations

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- KU24.** how to check the quality of machined components against the specified quality standards
- KU25.** use of HSS, Tungsten carbide, Ceramic and Diamond indexable tips, and factors which determine their selection and use
- KU26.** use of various work holding devices
- KU27.** importance of conducting cutting trial, methods of trial dry run, single block checks, cutting with offset adjustment to get oversize part
- KU28.** parameters to be checked before operating in auto mode dimensions, surface finishes
- KU29.** importance of periodic maintenance checks for the machine and what are the common maintenance checks
- KU30.** selection of cutting tools, tool materials, chip breaker geometry, selecting cutting parameters from tool catalogues, selecting coolant
- KU31.** extent of their own authority and to whom they should report if they have problems that they cannot resolve

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret correctly the job specifications from growing/ job card, manuals, safety instructions etc. in English and/ or local language
- GS2.** able to fill up the required formats/ documents in English and / or local language
- GS3.** interact and communicate with supervisor or other company personnel as per requirement
- GS4.** shall be able to use simple numerical computation such as addition, subtraction, multiplication, division, fractions and decimal, percentages and proportions, simple ratios and average
- GS5.** check and clarify task-related information
- GS6.** plan, prioritize and sequence work operations as per job requirements
- GS7.** shall be able to detect out of tolerance limit of component or any malfunctioning of the machine and take corrective action
- GS8.** shall be able to decide when to contact supervisor in case of any unresolved problems
- GS9.** shall be able to analyse and interpret geometric dimensions and tolerance and apply balanced judgments to different instructions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5	10	-	-
PC1. understand and comply with safety, environmental & other relevant regulations and guidelines	1	2	-	-
PC2. wear personal protective equipment (PPE) like safety glasses, apron, no loose cloths/ hair, safety shoes while performing milling operations	1	2	-	-
PC3. ensure work area is clean and safe	1	2	-	-
PC4. ensure that machine safety guards are in place and are in correctly working condition	1	2	-	-
PC5. ensure that all tools, equipments are in a safe and usable conditions	1	2	-	-
<i>Prerequisite for operating CNC milling machine</i>	11	22	-	-
PC6. ensure availability of job specification i.e. approved drawings, sketches, instructions from the supervisor, job instruction sheet/ job card	1	2	-	-
PC7. read and understand the Job requirements from the job specifications and attention shall be given to the geometric tolerances	1	2	-	-
PC8. check the work piece material for the dimensions and ensure that it is free from foreign objects, dirt or other contamination and is within the required size	1	2	-	-
PC9. plan to perform the machining operations and the sequence of operations as per required job specifications on CNC milling machine	1	2	-	-
PC10. obtain all the appropriate tools and measuring instruments/ gauges required for the job	1	2	-	-
PC11. check the CNC milling machine for its functioning and ensure that it is ready for operation	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. prepare the CNC milling machine for the operations by mounting and setting the required workholding & tool holder devices and cutting tools	1	2	-	-
PC13. clarify any doubt, if any and see necessary instruction /training on the operation of the CNC milling machine whenever required	1	2	-	-
PC14. hold the work piece securely and correctly, without distortion	1	2	-	-
PC15. adjust the CNC milling machine settings as per job requirement to maintain desired accuracy	1	2	-	-
PC16. perform daily maintenance of machine according to defined checklist, at the beginning of day's shifts	1	2	-	-
<i>Performing Operations on CNC milling machine</i>	30	62	-	-
PC17. use and extract information from engineering drawings, dimensioning and tolerances	1	2	-	-
PC18. use and extract information from reference charts, tables, graphs and Engineering standards	1	2	-	-
PC19. load and unload component(s) using pre-determined fixtures or work holding devices as per work instructions	1	2	-	-
PC20. make basic program and check correctness of program through dry run and single block check	1	2	-	-
PC21. adjust and set the speed and feed of the CNC milling machine to achieve the job specifications	1	3	-	-
PC22. operate the machine tool controls safely and correctly, in line with operational procedures	1	3	-	-
PC23. stop the CNC milling machine, both in normal and emergency situations correctly by following the right procedure and should be able to restart the machine after the emergency	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. do first part cutting trial by setting tool offsets to get oversize part	1	3	-	-
PC25. measure the critical parameters of the machined component on machine (without removing from the machine), after the trial run	1	3	-	-
PC26. correct the offsets based on the measurements by accessing program edit facility in order to enter tooling data	1	3	-	-
PC27. measure the component after unloading to check for accuracy in the critical parameters as per job specifications	1	3	-	-
PC28. produce machined components that combine different machining operations and have a range of features	1	3	-	-
PC29. follow the specified machining sequence and procedure as per job specifications	1	3	-	-
PC30. interpret in-built machine alarms and respond to the same as per operating manual or specified instructions	1	3	-	-
PC31. observe for inconsistency in dimensions due to tool wear and correct the offsets accordingly	2	3	-	-
PC32. ensure that machine settings are adjusted as and when required, either by self or the setter, to maintain the required accuracy	2	3	-	-
PC33. identify when tools need replacement and replace worn tool with new tool	2	3	-	-
PC34. produce components as per required standards	2	3	-	-
PC35. report problems and seek appropriate assistance in a timely manner	2	3	-	-
PC36. complete documentation during and post operations as per organizational procedures and applicable quality management system	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. return the machine and all tools and equipment to the correct location on completion of activities	2	3	-	-
PC38. leave the work area in a safe and tidy condition on completion of job activities as per 5S practices	2	3	-	-
<i>Seeking Guidance for unresolved Problems</i>	4	6	-	-
PC39. Report the problem to the supervisor, if it cannot be resolved	2	3	-	-
PC40. seek guidance from the supervisor/ specialist of the problem is outside his/her area of competence	2	3	-	-
NOS Total	50	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0802
NOS Name	Perform machining operations on metal or plastic workpieces using Computer Numerically Controlled Milling machines
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	CNC Milling, CNC Milling
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

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- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

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- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0405: To Effective working with others

Description

This unit covers basic etiquette and competencies that an individual is required to possess and demonstrate in their behavior and interactions with others at the workplace. This covers areas such as communication, discipline, handling conflict and grievances.

Scope

The scope covers the following :

- This unit/task covers the following:
- Effective Working with others

Elements and Performance Criteria

Effective working with others

To be competent, the user/individual on the job must be able to:

- PC1.** display appropriate communication etiquette while working
- PC2.** display active listening skills while interacting with others at work
- PC3.** demonstrate responsible & disciplined behaviors at workplace
- PC4.** accurately receive information & instructions from supervisor & fellow workers, getting clarification where required
- PC5.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC6.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC7.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC8.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures followed in the company for working with others in an organizational set-up
- KU2.** grievance/ conflict handling mechanism of the company
- KU3.** relevant people and their responsibilities within the work area
- KU4.** importance of effective communication in the workplace
- KU5.** importance of teamwork in organizational and individual success
- KU6.** barriers to effective communication



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- KU7.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU8.** various categories of people that one is required to communicate and co-ordinate within the organization
- KU9.** importance of discipline for professional success
- KU10.** importance of ethics for professional success
- KU11.** what constitutes disciplined behavior for a working professional
- KU12.** common reasons for interpersonal conflict
- KU13.** importance and ways of managing interpersonal conflict effectively
- KU14.** importance of developing effective working relationships for professional success
- KU15.** expressing and addressing grievances appropriately and effectively

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective working with others</i>	10	10	-	-
PC1. display appropriate communication etiquette while working	1	1	-	-
PC2. display active listening skills while interacting with others at work	1	1	-	-
PC3. demonstrate responsible & disciplined behaviors at workplace	1	1	-	-
PC4. accurately receive information & instructions from supervisor & fellow workers, getting clarification where required	1	1	-	-
PC5. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	2	2	-	-
PC6. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	2	2	-	-
PC7. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	1	1	-	-
PC8. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	1	1	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0405
NOS Name	To Effective working with others
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025



Qualification Pack

CPC/CAP/N0404: Computer and data entry in MS Office/Office Open Source software

Description

This NOS covers the skills and knowledge required to perform data entry operations and prepare official documents using MS Office or open-source office applications such as LibreOffice or OpenOffice.

Scope

The scope covers the following :

- It includes entering, updating, and managing data accurately, as well as creating, formatting, and presenting documents, spreadsheets, and presentations for organizational use.

Elements and Performance Criteria

Enter, update and maintain data

To be competent, the user/individual on the job must be able to:

- PC1.** Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents in to Computer application having MS OFFICE software
- PC2.** Scan source documents in accordance with specific instructions
- PC3.** Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data
- PC4.** Maintain files of source documents or other information related to data entered
- PC5.**
 - Investigate and confirm data that is unclear before entering, generate reports of data entry, store
 - completed work in designated locations and perform backup operations
- PC6.** Update database information to reflect most current source information
- PC7.** Assist in the filing and storage of security and back up data files
- PC8.** Respond to requests for information and access relevant files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The data management applications/tools used by the company
- KU2.** Data entry protocol
- KU3.** Data integrity and security policies of the company
- KU4.** Approved methods for carrying document control and archiving
- KU5.** Basic understanding computers and its terminology
- KU6.** Work on MS office software

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Efficiently enter data into computer applications
- GS2.** Prepare legible reports
- GS3.** Read and understand manuals, SOPs, instructions, memos, reports, job cards etc
- GS4.** Communicate effectively with the team members and supervisors
- GS5.** Apply basic logic to identify data errors
- GS6.** Pay attention to details
- GS7.**
 - Plan assigned tasks within timeline and as per priority order
 - specified
- GS8.** Identify process improvements

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Enter, update and maintain data</i>	10	10	-	-
PC1. Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents in to Computer application having MS OFFICE software	1	1	-	-
PC2. Scan source documents in accordance with specific instructions	1	1	-	-
PC3. Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data	1	1	-	-
PC4. Maintain files of source documents or other information related to data entered	1	1	-	-
PC5. - Investigate and confirm data that is unclear before entering, generate reports of data entry, store - completed work in designated locations and perform backup operations	2	1	-	-
PC6. Update database information to reflect most current source information	2	2	-	-
PC7. Assist in the filing and storage of security and back up data files	1	2	-	-
PC8. Respond to requests for information and access relevant files	1	1	-	-
NOS Total	10	10	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0404
NOS Name	Computer and data entry in MS Office/Office Open Source software
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

Qualification Pack

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** read and write different types of documents/instructions/correspondence
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Career Development & Goal Setting</i>	1	2	-	-
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches are assigned to Training Assessment Wing (TAW), CIPET HO for planning of assessment
- Training Centers request TAW for Assessment and Certification of Trainees
- TAW identifies suitable assessor and nominates the assessor to the respective Training Centre
- TAW monitors the assessment process
- Training Centers maintains necessary records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 02 Assessors in a day (or) 01 assessor in 2 days

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- Check that the allotted time to the candidates to complete the Theory & Practical Assessment

3. Assessment Quality Assurance levels/Framework:

- Question bank / Question Paper is prepared by the Subject Matter Experts (SME) / Assessor
- Questions are mapped to the specified assessment criteria
- Certified Assessor & Trainer will be engaged in the process

4. Types of evidence or evidence-gathering protocol:

- Assessment batch - Group Photo of Trainees along with Assessor

5. Method of verification or validation:

- Surprise visit to the assessment location
- Date / Time recorded for the reporting of the assessor from assessment location
- Virtual meet with the Assessor / Trainees

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0801.Perform machining operations on metal or plastic material using Conventional Milling machine	50	80	0	0	130	20
CPC/CAP/N0803.Programming of Computer Numerically Controlled (CNC) Milling Machines	50	150	0	0	200	20

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0802.Perform machining operations on metal or plastic workpieces using Computer Numerically Controlled Milling machines	50	100	0	0	150	20
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	-	-	40	10
CPC/CAP/N0405.To Effective working with others	10	10	0	0	20	10
CPC/CAP/N0404.Computer and data entry in MS Office/Office Open Source software	10	10	0	0	20	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	410	-	-	610	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.